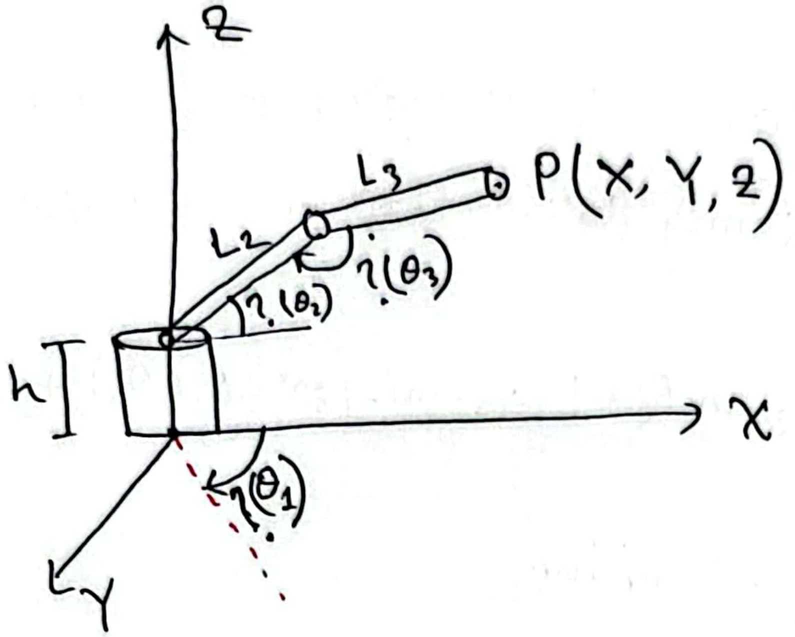
Inverse Kinematics

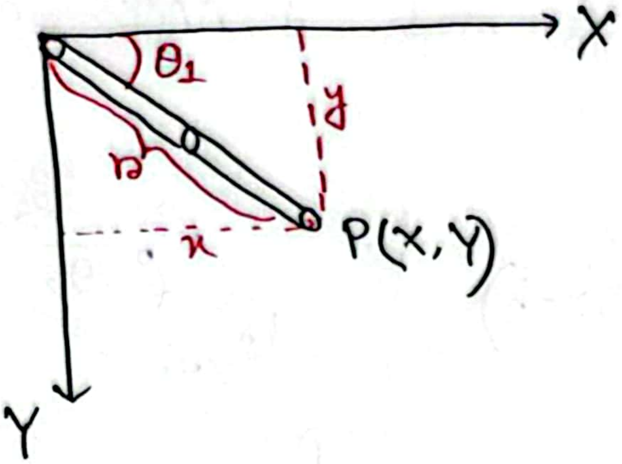
$$(X, Y, Z) \longrightarrow (\theta_1, \theta_2, \theta_3)$$



From this Question we can not visualize the whole X, Y, Z in our 2D geometry approach. Therefore, we will use two plans two solve this part.

First top view and then side view.
Why ? do it yourself! ☺

Top view:

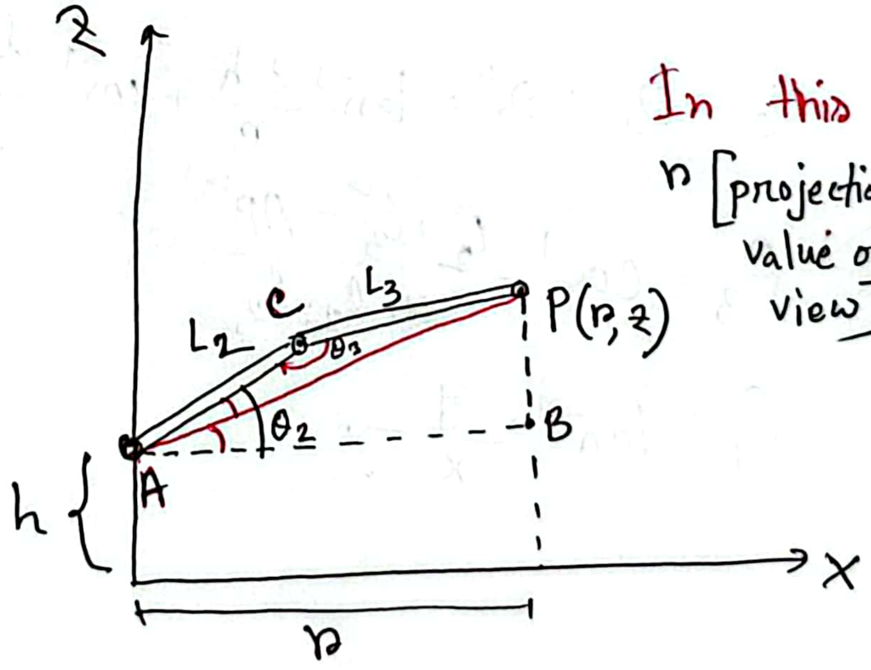


$\tan\theta_1 = \frac{y}{x} \implies \theta_1 = \tan^{-1}\frac{y}{x}$ [Ans]

Also, we need the value of r when ~~to~~ shift to the side view.

$$r = \sqrt{x^2 + y^2}$$

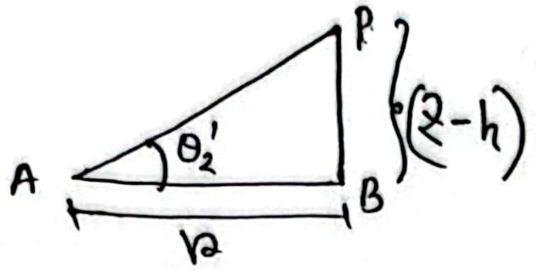
Side View:



In this figure basically r [projection of arm becom value of x- in side view].

θ_2 divided into two parts.

$\theta_2 = \Delta APB$'s one angle $+$ ΔACP's one side angle

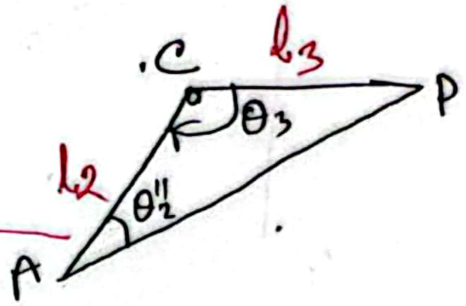


$$\theta_2' = \tan^{-1}\frac{z-h}{r}$$

$$= \tan^{-1}\frac{z-h}{\sqrt{x^2+y^2}}$$

$$AP^2 = r^2 + (z-h)^2$$

$$= \left(\sqrt{x^2+y^2}\right)^2 + (z-h)^2$$



This is not right triangle. That's why we use cosine rule

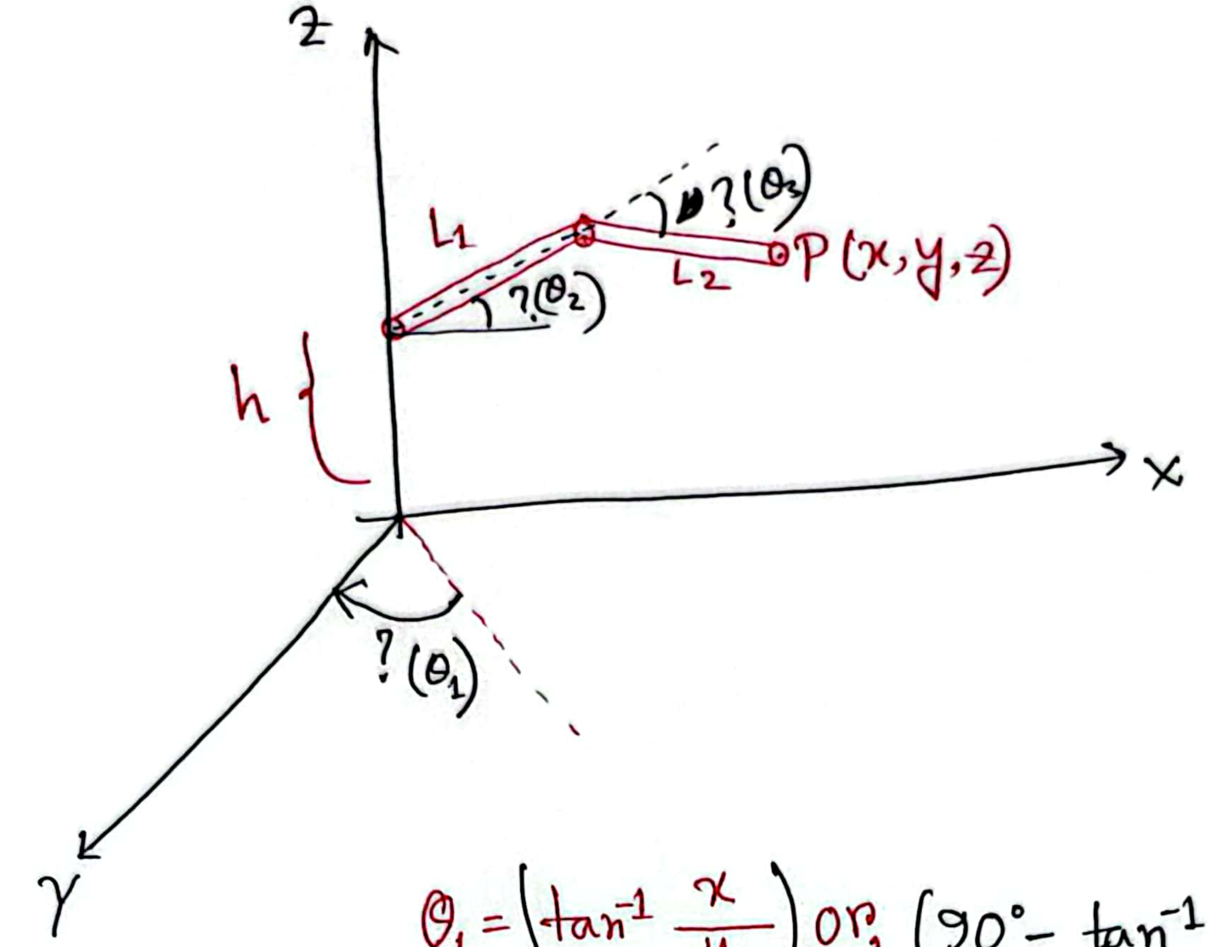
$$\cos\theta'' = \frac{l_2^2 + AP^2 - l_3^2}{2 \cdot l_2 \cdot AP}$$

$$\therefore \theta_2 = \theta_2' + \theta_2'' = \tan^{-1}\frac{z-h}{r} + \cos^{-1}\frac{l_2^2 + AP^2 - l_3^2}{2 \cdot l_2 \cdot AP}$$

$$\theta_3 = \cos^{-1}\frac{l_2^2 + l_3^2 - AP^2}{2 \cdot l_2 \cdot l_3}$$

$$\theta_1 = \tan^{-1}\frac{y}{x}.$$

Extension:



$$\theta_1 = \left(\tan^{-1}\frac{x}{y}\right) \text{ or, } \left(90^\circ - \tan^{-1}\frac{y}{x}\right)$$

$$\theta_2 = \tan^{-1}\frac{z-h}{\sqrt{x^2+y^2}} + \cos^{-1}\frac{l_2^2 + (x^2+y^2)+(z-h)^2 - l_3^2}{2 \cdot l_2 \cdot \sqrt{(x^2+y^2)+(z-h)^2}}$$

$$\theta_3 = 180^\circ - \cos^{-1}\frac{l_2^2 + l_3^2 - \left(\sqrt{(x^2+y^2)+(z-h)^2}\right)^2}{2 \cdot l_2 \cdot l_3}$$

[You can do this yourself]